JCEI'S JAIHIND COLLEGE OF ENGINEERING, KURAN

A.Y. 2025-26

MINI PROJECT ON

"BANKING SYSTEM''

UNDER

"OBJECT ORIENTED PROGRAMMING"

SE-AI&DS ENGINEERING

DEPARTMENT OF AI & DS ENGINEERING

SEMESTER-IV

SUBMITTED BY

NAME OF STUDENT – Shete Priti Rahul

Department of Artificial Intelligence and data science Engineering

(2025-26)

Certificate

This is to certify that,

NAME OF STUDENT – Shete Priti Rahul

I have successfully completed the Mini project entitled **"Banking System"** Under the guidance of **Prof.R. V. Varhade** in partial fulfillment of the requirements for the Second Year of Engineering in Artificial intelligence and Data Science under the Savitribai Phule Pune University during the academic year 2025-2026.

Prof. R. V.Varhade

Subject Teacher

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Yours faithfully,
Ms Shete Priti Rahul

ABSTRACT

The Banking System Mini Project is a console-based application developed using Java, JDBC, and MySQL to simulate basic banking operations in a simple and efficient manner. The main objective of this system is to manage bank account activities such as creating an account, depositing money, withdrawing money, checking balance, and displaying account details. Java is used to implement the application logic and handle user interaction, while MySQL is used as the backend database to store and manage all customer and transaction data securely. JDBC acts as a bridge between Java and MySQL, enabling smooth communication between the application and the database. Exception handling is also incorporated to manage runtime errors and ensure the system runs reliably even in case of invalid inputs or unexpected situations. This project helps in understanding real-time banking processes and strengthens core concepts of object-oriented programming, database connectivity, and error handling. It also demonstrates how automation can improve efficiency, accuracy, and reliability in traditional banking systems, making it a useful learning project for software development fundamentals.

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1. INTRODUCTION

In the modern digital world, banking systems have become an essential part of everyday life, enabling individuals and organizations to manage financial transactions efficiently and securely. Traditional banking methods involved manual record-keeping, which was time-consuming, prone to human errors, and difficult to maintain. With the advancement of technology, computerized banking systems have replaced these traditional methods, providing faster, more reliable, and accurate services. This mini project, titled *Banking System*, is developed to simulate the basic functionalities of a real-world banking application using Java programming, JDBC (Java Database Connectivity), and MySQL database.

The primary aim of this project is to design and implement a simple console-based banking system that allows users to perform core banking operations such as account creation, depositing money, withdrawing money, checking account balance, and displaying account information. In addition to these features, the system also enforces a daily withdrawal limit, which adds a layer of control and reflects real banking constraints. This makes the project not only functional but also closer to real-life banking scenarios.

The project utilizes Java as the programming language due to its platform independence, object-oriented features, and robustness. JDBC is used to establish a connection between the Java application and the MySQL database, enabling seamless data storage and retrieval. MySQL serves as the backend database where all account-related information, such as account number, customer name, balance, and transaction limits, is stored and managed efficiently. The interaction between Java and the database demonstrates how real-world applications handle persistent data.

Another significant aspect of this project is the implementation of exception handling mechanisms in Java. Exception handling plays a crucial role in building reliable and user-friendly applications by preventing unexpected program termination. In this system, various exception handling concepts such as try-catch blocks, multiple catch statements, nested try statements, the use of throw and throws keywords, and the finally block are incorporated. These techniques help in managing runtime errors like invalid input, database connection failures, insufficient balance, and exceeding withdrawal limits, thereby improving the stability of the application.

Furthermore, this project helps in understanding the practical implementation of database operations such as inserting, updating, and retrieving data using SQL queries. It also introduces students to real-time problem-solving, logical thinking, and structured programming practices. The modular design of the system ensures that each functionality is handled separately, making the code easier to understand, maintain, and extend in the future.

In conclusion, the Banking System Mini Project serves as a foundational application that integrates programming concepts with database management and error handling techniques. It provides valuable hands-on experience in developing real-world software applications and prepares students for more advanced projects in the field of software engineering and application development.

2. OBJECTIVES

1. To design and implement a console-based Banking System application using Java.
2. To establish a connection between Java application and MySQL database using JDBC.
3. To understand and implement database connectivity concepts such as DriverManager, Connection, Statement, and ResultSet.
4. To perform CRUD operations (Create, Read, Update, Delete) on banking records efficiently.
5. To develop core banking functionalities:
   - Account creation
   - Deposit operation
   - Withdrawal operation
   - Balance inquiry
   - Display account details
6. To implement daily withdrawal limit logic for secure transaction handling.
7. To apply Exception Handling techniques including:
   - try-catch blocks
   - multiple catch clauses
   - nested try statements
   - use of throw and throws
   - finally block for resource management
8. To handle built-in exceptions such as SQLException, ArithmeticException, and NullPointerException.
9. To ensure data validation for user inputs (e.g., invalid amount, insufficient balance).
10. To maintain data integrity and consistency during database transactions.
11. To understand and implement transaction management (commit and rollback).
12. To improve error handling and debugging skills in Java applications.
13. To apply Object-Oriented Programming (OOP) concepts like classes, objects, encapsulation, and methods.
14. To develop a modular and structured code design for better readability and maintenance.
15. To enhance knowledge of real-world banking operations and workflows.
16. To gain practical exposure to backend development using Java and databases.
17. To build a foundation for developing advanced enterprise-level applications.
18. To improve logical thinking and problem-solving skills through implementation.
19. To understand the importance of secure data handling in financial systems.
20. To prepare for real-world projects and technical interviews related to Java and JDBC.

3. REQUIREMENT SPECIFICATION

1. **Hardware Requirements**

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: At least 500 MB free space
- System Type: 32-bit or 64-bit computer

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2. **Software Requirements**

- Operating System: Windows 10/11 (or Linux/Mac)
- Programming Language: Java (JDK 8 or above)
- IDE: Eclipse IDE
- Database: MySQL Server
- Database Tool: MySQL Command Line / MySQL Workbench
- JDBC Driver: MySQL Connector/J
- Browser: Any web browser (for documentation/reference)

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3. **Functional Requirements**

The system should be able to:

- Create a new bank account with user details
- Store account information in the database
- Deposit money into an account
- Withdraw money from an account
- Check account balance
- Display account details
- Enforce daily withdrawal limit
- Validate user inputs (e.g., negative amount not allowed)
- Handle invalid account numbers
- Provide proper error messages for failed operations

---

4. **Non-Functional Requirements**

- Performance: System should respond quickly to user inputs
- Reliability: Transactions should be accurate and consistent
- Security: Basic validation to prevent unauthorized access

- Usability: Easy-to-use console interface
- Maintainability: Code should be well-structured and readable
- Scalability: Can be extended with more features in future

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5. Database Requirements

- Database Name: OOP_priti
- Table: accounts

Fields in Table:

- Account Number (Primary Key)
- Account Holder Name
- Balance
- Daily Withdrawal Limit

---

6. System Constraints

- Works on local system (no online deployment)
- Limited to basic banking operations only
- No GUI (console-based application)
- Requires MySQL server to be running

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7. Assumptions

- User has basic knowledge of operating the system
- Database connection is properly configured
- JDBC driver is correctly added to the project
- Valid inputs are provided by the user

4. WORKFLOW

1. Start the application. If the database connection is not established, the system cannot proceed.
2. Load JDBC driver and connect to MySQL database.
3. Display the main menu with available operations.
4. Accept user input. Only valid choices are considered—invalid input leads to an error message.
5. Check the selected option and proceed accordingly:

Step 5.1 – Create Account

- Enter required details.
- Validate input.
- Insert data into database.

Step 5.2 – Deposit Money

- Enter account number and amount.
- Validate amount.
- Update balance in database.

Step 5.3 – Withdraw Money

- Enter account number and amount.
- Check sufficient balance and daily limit.
- Deduct amount if valid.

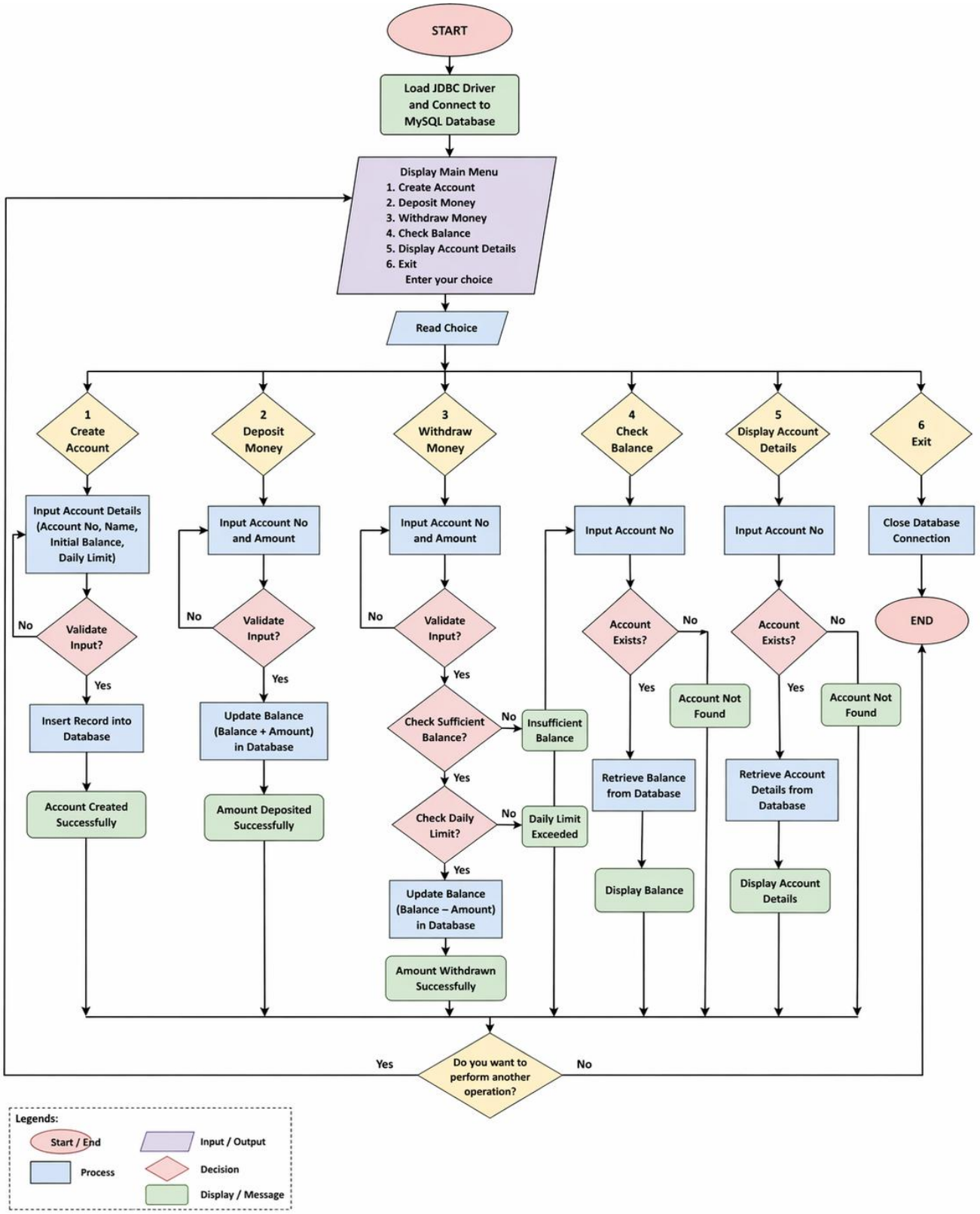
Step 5.4 – Check Balance

- Enter account number.
- Retrieve balance from database.

Step 5.5 – Display Account Details

- Enter account number.
- Fetch and display all details.

6. Handle errors using exception handling (try-catch). Invalid operations are not executed.
7. After completing the operation, return to the main menu.
8. Repeat steps until the user selects Exit.
9. Close database connection properly.
10. End the application.


START
Load JDBC Driver and Connect to MySQL Database
Display Main Menu
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Display Account Details
6. Exit
Enter your choice
Read Choice
1 Create Account
2 Deposit Money
3 Withdraw Money
4 Check Balance
5 Display Account Details
6 Exit
Input Account Details (Account No, Name, Initial Balance, Daily Limit)
Validate Input?
No
Yes
Insert Record into Database
Account Created Successfully
Input Account No and Amount
Validate Input?
No
Yes
Update Balance (Balance + Amount) in Database
Amount Deposited Successfully
Input Account No and Amount
Validate Input?
No
Yes
Check Sufficient Balance?
No
Insufficient Balance
Yes
Check Daily Limit?
No
Daily Limit Exceeded
Yes
Update Balance (Balance – Amount) in Database
Amount Withdrawn Successfully
Input Account No
Account Exists?
No
Account Not Found
Yes
Retrieve Balance from Database
Display Balance
Input Account No
Account Exists?
No
Account Not Found
Yes
Retrieve Account Details from Database
Display Account Details
Close Database Connection
END
Do you want to perform another operation?
Yes
No
Legends:
Start / End
Process
Input / Output
Decision
Display / Message

5. FUNCTIONALITY

The Banking System provides the following core functionalities:

1. Account Creation

- Allows the user to create a new bank account.
- Stores details such as account number, name, balance, and daily withdrawal limit in the database.
- Ensures that duplicate account numbers are not allowed.

2. Deposit Money

- Enables the user to deposit money into an existing account.
- Updates the account balance in the database.
- Rejects invalid or negative amounts.

3. Withdraw Money

- Allows the user to withdraw money from their account.
- Checks for:
    - Sufficient balance
    - Daily withdrawal limit
- Prevents withdrawal if conditions are not satisfied.
- Updates balance after successful transaction.

4. Balance Inquiry

- Displays the current balance of a specific account.
- Fetches real-time data from the database.

5. Display Account Details

- Shows complete account information including:
    - Account number
    - Account holder name
    - Current balance
    - Withdrawal limit

6. Input Validation

- Ensures correct and valid data entry.
- Handles invalid account numbers and incorrect inputs.

7. Exception Handling

- Uses try-catch blocks to handle runtime errors.
- Manages database errors (SQLException).
- Prevents system crash due to invalid operations.

8. Database Connectivity

- Connects Java application with MySQL using JDBC.
- Executes SQL queries for all operations.

9. Transaction Management

- Ensures safe and consistent database operations.
- Uses commit and rollback when required.

10. Menu-Driven Interface

- Provides a simple console-based menu for user interaction.
- Allows repeated operations until the user exits.

6. CONCEPTUAL DESIGN

The conceptual design represents the overall structure of the Online Voting System. It identifies themain entities, their attributes, and the relationships between them without focusing on implementation details. This design helps in understanding how data flows and interacts within the system.

1. System Overview

The Banking System is a **console-based application** developed using Java and JDBC, connected to a MySQL database. It allows users to perform basic banking operations such as account creation, deposit, withdrawal, and balance inquiry.

2. Main Components

a. User Interface (UI)

- Console-based interface
- Accepts user input through keyboard
- Displays menu and results

b. Application Layer (Java Logic)

- Contains business logic for all operations
- Validates user inputs
- Handles transaction processing
- Implements exception handling

c. Database Layer (MySQL)

- Stores all account-related data
- Performs CRUD operations
- Ensures data consistency

3. Entities and Attributes

Entity: Account

- Account Number (Primary Key)
- Account Holder Name
- Balance
- Daily Withdrawal Limit

4. Relationships

- One user is associated with one account (in this basic system).
- The application interacts with the database to perform operations on the Account entity.

5. Data Flow (Conceptual)

- User inputs data →
- Application processes and validates →
- JDBC sends SQL queries →
- Database executes queries →
- Results returned to application →

- Output displayed to user

6. Functional Modules

- **Account Management Module**
  Handles account creation and storage
- **Transaction Module**
  Handles deposit and withdrawal operations
- **Inquiry Module**
  Handles balance check and account details display
- **Exception Handling Module**
  Handles errors and ensures smooth execution

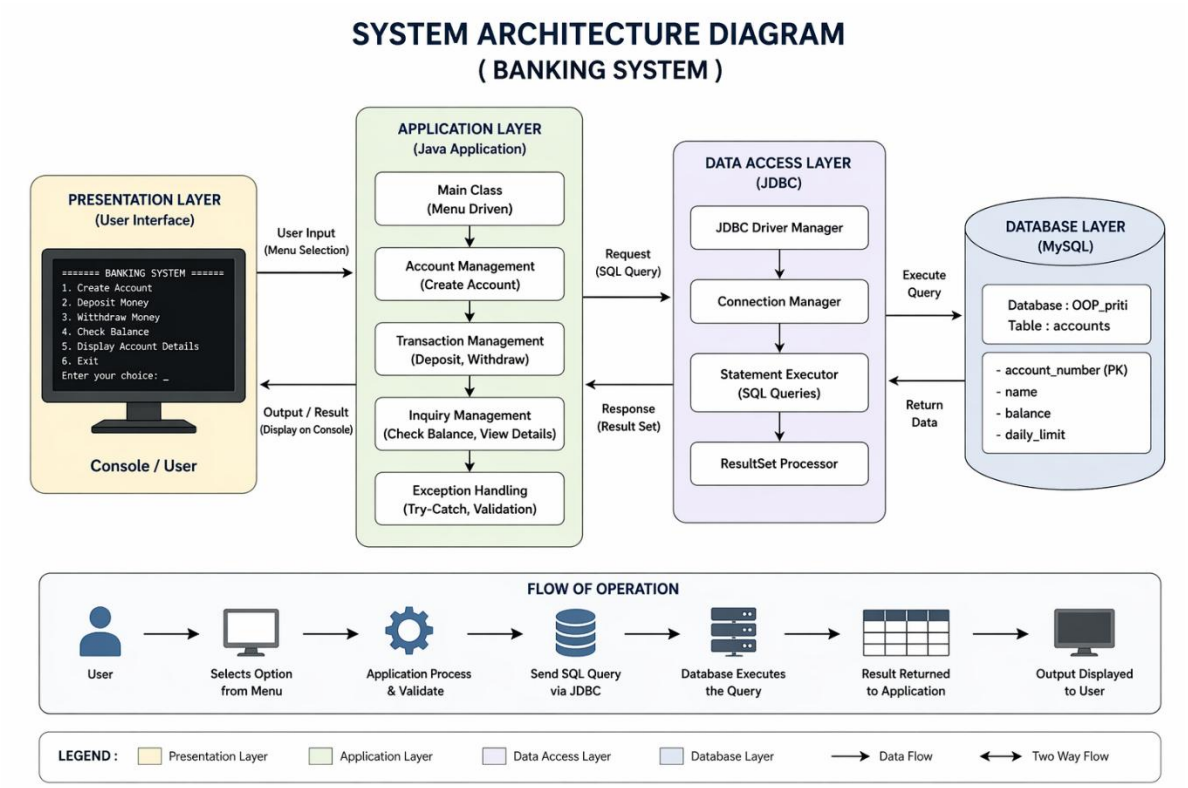
7. Design Approach

- Follows **Object-Oriented Programming (OOP)** principles
- Uses **modular design** for better maintainability
- Ensures **separation of concerns** between UI, logic, and database

8. Constraints in Design

- Console-based (no graphical interface)
- Single-user system
- Limited to basic banking features

System Architecture diagram



7. IMPLEMENTATION

1. JAVACode

```
package bankingsystem;
import java.sql.*;
import java.util.Scanner;

public class BankingSystem {

    static final String URL = "jdbc:mysql://localhost:3306/OOP_priti";
    static final String USER = "root";
    static final String PASSWORD = "Priti@123";

    static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }

    // Create Account
    static void createAccount(String name, double balance) {
        try (Connection con = getConnection()) {
            String sql = "INSERT INTO accounts(name, balance, daily_limit) VALUES (?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, name);
            ps.setDouble(2, balance);
            ps.setDouble(3, 5000); // daily limit
            ps.executeUpdate();
            System.out.println("Account Created Successfully!");
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Deposit
    static void deposit(int id, double amount) {
        try (Connection con = getConnection()) {
            String sql = "UPDATE accounts SET balance = balance + ? WHERE acc_id = ?";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setDouble(1, amount);
            ps.setInt(2, id);
            ps.executeUpdate();
            System.out.println("Deposit Successful!");
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Withdraw
    static void withdraw(int id, double amount) {
        try (Connection con = getConnection()) {

            String check = "SELECT balance, daily_limit FROM accounts WHERE acc_id=?";
            PreparedStatement ps1 = con.prepareStatement(check);
            ps1.setInt(1, id);
            ResultSet rs = ps1.executeQuery();

            if (rs.next()) {
                double balance = rs.getDouble(1);
                double limit = rs.getDouble(2);

```

```
                if (amount > balance) {
                    System.out.println("Insufficient Balance!");
                } else if (amount > limit) {
                    System.out.println("Daily limit exceeded!");
                } else {
                    String sql = "UPDATE accounts SET balance = balance - ? WHERE acc_id = ?";
                    PreparedStatement ps2 = con.prepareStatement(sql);
                    ps2.setDouble(1, amount);
                    ps2.setInt(2, id);
                    ps2.executeUpdate();
                    System.out.println("Withdrawal Successful!");
                }
            }

        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Check Balance
    static void checkBalance(int id) {
        try (Connection con = getConnection()) {
            String sql = "SELECT balance FROM accounts WHERE acc_id=?";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setInt(1, id);
            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                System.out.println("Balance: " + rs.getDouble(1));
            }
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Display Account
    static void display(int id) {
        try (Connection con = getConnection()) {
            String sql = "SELECT * FROM accounts WHERE acc_id=?";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setInt(1, id);
            ResultSet rs = ps.executeQuery();

                if (rs.next()) {
                    System.out.println("ID: " + rs.getInt(1));
                    System.out.println("Name: " + rs.getString(2));
                    System.out.println("Balance: " + rs.getDouble(3));
                    System.out.println("Daily Limit: " + rs.getDouble(4));
                }
            } catch (Exception e) {
                System.out.println(e);
            }
        }

        public static void main(String[] args) {

            Scanner sc = new Scanner(System.in);

            while (true) {
                System.out.println("\n--- BANK MENU ---");
```

```
                    System.out.println("1. Create Account");
                    System.out.println("2. Deposit");
                    System.out.println("3. Withdraw");
                    System.out.println("4. Check Balance");
                    System.out.println("5. Display Account");
                    System.out.println("6. Exit");

                    System.out.print("Enter choice: ");
                    int choice = sc.nextInt();

                    switch (choice) {

                        case 1:
                            System.out.print("Enter Name: ");
                            String name = sc.next();
                            System.out.print("Enter Initial Balance: ");
                            double bal = sc.nextDouble();
                            createAccount(name, bal);
                            break;

                        case 2:
                            System.out.print("Enter Account ID: ");
                            int id1 = sc.nextInt();
                            System.out.print("Enter Amount: ");
                            double dep = sc.nextDouble();
                            deposit(id1, dep);
                            break;

                        case 3:
                            System.out.print("Enter Account ID: ");
                            int id2 = sc.nextInt();
                            System.out.print("Enter Amount: ");
                            double wd = sc.nextDouble();
                            withdraw(id2, wd);
                            break;

                        case 4:
                            System.out.print("Enter Account ID: ");
                            int id3 = sc.nextInt();
                            checkBalance(id3);
                            break;

                        case 5:
                            System.out.print("Enter Account ID: ");
                            int id4 = sc.nextInt();
                            display(id4);
                            break;

                        case 6:
                            System.exit(0);

                        default:
                            System.out.println("Invalid Choice!");
                    }
                }
            }
        }
```

OUTPUT:

eclipse-workspace - miniproject/src/bankingsystem/BankingSystem.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Console
Eclipse IDE for Java Developers 2026-03 Release
BankingSystem (1) [Java Application] C:\Users\shete\.p2\pool\plugins\org.eclipse.justj.
--- BANK MENU ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Display Account
6. Exit
Enter choice:
1
Enter Name: alex
Enter Initial Balance: 500
Account Created Successfully!
--- BANK MENU ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Display Account
6. Exit
Enter choice: 2
Enter Account ID: 1
Enter Amount: 20
Deposit Successful!
--- BANK MENU ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Display Account
6. Exit
Enter choice: 4
Enter Account ID: 1
Balance: 540.0
--- BANK MENU ---
1. Create Account
2. Deposit
3. Withdraw

2.MySQL

```
CREATE DATABASE OOP_priti;
USE OOP_priti;


CREATE TABLE accounts
(
acc_id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
balance DOUBLE,
daily_limit DOUBLE
);


SHOW TABLES;
DESCRIBE accounts;
```

8. TESTING

The testing phase confirmed that the system is reliable, secure, and efficient. It performs all operations correctly and meets the required objectives. Any minor errors found during testing were corrected, ensuring smooth system performance.

```
MySQL 8.0 Command Line Cli

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE OOP_priti;
Query OK, 1 row affected (0.02 sec)

mysql> USE OOP_priti;
Database changed
mysql> CREATE TABLE accounts (acc_id INT PRIMARY KEY AUTO_INCREMENT,name VARCHAR(100),balance DOUBLE,daily_limit DOUBLE);
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+---------------------+
| Tables_in_oop_priti |
+---------------------+
| accounts            |
+---------------------+
1 row in set (0.04 sec)

mysql> DESCRIBE accounts;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| acc_id      | int          | NO   | PRI | NULL    | auto_increment |
| name        | varchar(100) | YES  |     | NULL    |                |
| balance     | double       | YES  |     | NULL    |                |
| daily_limit | double       | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)

mysql> DESCRIBE accounts;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| acc_id      | int          | NO   | PRI | NULL    | auto_increment |
| name        | varchar(100) | YES  |     | NULL    |                |
| balance     | double       | YES  |     | NULL    |                |
| daily_limit | double       | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)
```

9. CONCLUSION

The Banking System mini project successfully demonstrates the implementation of a simple banking application using Java, JDBC, and MySQL. The system is capable of performing essential banking operations such as account creation, deposit, withdrawal, balance inquiry, and displaying account details in an efficient and structured manner.

Through this project, the integration between Java and the MySQL database using JDBC has been effectively achieved, ensuring smooth data storage and retrieval. The use of exception handling techniques has improved the reliability of the system by preventing crashes and handling runtime errors properly.

The project also highlights the application of Object-Oriented Programming concepts and modular design, making the system easy to understand, maintain, and extend. Proper validation and transaction handling ensure data consistency and accuracy during operations.

Although the system is limited to a console-based interface and basic functionalities, it provides a strong foundation for developing more advanced banking applications with graphical interfaces, enhanced security, and multi-user support.

Overall, this project has enhanced practical knowledge of Java programming, database connectivity, and real-world application development, making it a valuable learning experience.

10. FUTURE SCOPE

The banking system project can be further enhanced in the future with more advanced features and technologies to make it more realistic and industry-ready.

1. Online Banking Integration

The system can be extended to support web or mobile banking so users can access accounts anytime from anywhere.

2. Mobile Application Development

A dedicated Android/iOS app can be developed for easy account management, transactions, and notifications.

3. Security Enhancements

Advanced security features like OTP verification, biometric authentication, and two-factor authentication (2FA) can be added.

4. Real-Time Transactions

Integration with payment gateways (UPI, net banking, debit/credit cards) to support real-time money transfers.

5. Loan and Credit Management

Modules for applying loans, checking eligibility, EMI calculation, and repayment tracking can be added.

6. AI-Based Fraud Detection

Machine learning can be used to detect suspicious transactions and prevent fraud.

7. Cloud Database Integration

Moving from local MySQL to cloud databases for better scalability and remote access.

8. Chatbot Support System

An AI chatbot can be integrated to assist users with banking queries and services.

9. Report Generation System

Advanced reporting tools for generating account statements, transaction history, and analytics.

10. Multi-Bank Support System

The system can be extended to handle multiple banks under one platform.

11. REFERENCES

1. **Java: The Complete Reference** – Herbert Schildt
   (Core Java concepts, OOP, exception handling, JDBC basics)

2. **Head First Java (2nd Edition)** – Kathy Sierra, Bert Bates
   (Easy understanding of Java fundamentals and programming logic)

3. **MySQL Documentation (Official)**
   https://dev.mysql.com/doc/
   (SQL commands, database creation, queries, constraints)

4. **Oracle Java Documentation (JDBC API)**
   https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/
   (JDBC connection, DriverManager, Connection, Statement, ResultSet)

5. **GeeksforGeeks – Java & JDBC Tutorials**
   https://www.geeksforgeeks.org/java/
   (Practical examples for CRUD operations and exception handling)

6. **W3Schools – SQL Tutorial**
   https://www.w3schools.com/sql/
   (Basic SQL queries: SELECT, INSERT, UPDATE, DELETE)

7. **YouTube Educational Channels (Java + MySQL Projects)**
   (For understanding mini project implementation and debugging)===